

Long-window (128-frame) world-space comparison on HOI4D

All metrics in mm, lower is better, over 157 held-out sequences scored in 128-frame segments. *Every* row uses the same scorer, the same segment set and `--drop.partial.tail`, so no method is scored on windows another never predicted. C-MPJPE_{abs} = absolute camera-frame joint error; C-MPJPE_{rr} = root-relative; WA-MPJPE = per-window Procrustes (30-frame and 128-frame windows); W-MPJPE = global error after a single first-window rigid alignment with *no* rescaling.

Method	Regime	Camera frame		Window-aligned		Global
		abs ↓	root-rel ↓	30f ↓	128f ↓	W-MPJPE ↓
Ours (feedforward)	online, no SLAM	35.2	26.6	35.0	64.0	200.9
Ours + DROID-SLAM ^a	offline + SLAM	34.4	26.5	36.0	63.9	201.7
HaWoR ^b	offline + SLAM	86.9	32.3	42.0	71.7	215.6
WiLoR + SLAM ^c	offline + SLAM	84.0	27.2	36.4	65.1	219.0
HaMeR + SLAM ^c	offline + SLAM	89.0	30.3	37.5	66.5	227.6
HaPTIC + SLAM ^c	offline + SLAM	153.6	29.3	37.8	66.9	239.4
Dyn-HaMR ^d	offline optim.	1327.4	59.7	49.9	84.2	276.6
<i>Oracle ceilings: what a perfect camera trajectory would buy</i>						
Ours + GT camera track ^e	oracle	33.0	24.9	18.0	23.9	50.5
+ GT rotation only ^f	oracle	35.2	–	35.2	70.6	166.2
+ GT translation only ^f	oracle	35.2	–	28.7	67.5	123.9
+ GT rotation <i>and</i> translation ^f	oracle	35.2	–	19.8	25.5	61.5

The headline of this table is the gap between the two blocks. Our camera-frame error (C-MPJPE_{abs} 35.2) is essentially unchanged by the oracle (33.0), and our 30-frame window-aligned error is already 35.0 mm, yet global W-MPJPE is 200.9. Handing the *same* per-frame hand predictions a ground-truth camera trajectory collapses W-MPJPE to 61.5 on the identical 157 sequences. Long-window global error is therefore dominated by camera-trajectory drift, not by hand pose: every method in the upper block sits within 38% of every other (200.9–276.6), a spread far smaller than the 3.3× drop the oracle delivers. This is why we report W-MPJPE alongside window-aligned metrics rather than instead of them.

^a Ours + DROID-SLAM: our per-frame camera-space hands composed with the same offline DROID-SLAM trajectory used by the baseline rows. It does *not* beat our own online predicted extrinsics (201.7 vs 200.9), i.e. replacing our camera head with offline SLAM buys nothing at this window length. ^b HaWoR: full pipeline (DROID-SLAM + Metric3D + motion + infiller), run end-to-end, scored identically. ^c WiLoR, HaMeR and HaPTIC do no camera tracking, so each is composed with the *same* DROID-SLAM trajectory and scored identically. None produces metric depth, so their absolute camera-frame column reflects a fitted rather than a predicted scale and should not be read as a metric-depth result; their root-relative and window-aligned columns are the meaningful ones. ^d Dyn-HaMR is a per-sequence test-time optimization method. In the input-matched setup its own metric-depth stage is bypassed, so world scale is unconstrained and absolute placement diverges (C-MPJPE_{abs} 1327); only its scale-invariant columns are meaningful. 156 of 157 sequences. ^e Reported over the 60-sequence subset for which a ground-truth camera track was available at the time of the run (120 segments), so it is *not* directly comparable cell-for-cell with the 157-sequence rows above. The full-157 equivalent is the last oracle row (61.5). ^f Trajectory-error decomposition on all 157 sequences: our own chained trajectory with the rotation, the translation, or both replaced by ground truth, hand predictions held fixed. Because only the camera changes, the camera-frame column is unchanged by construction (35.2); root-relative is omitted for the same reason. Translation is the larger of the two error sources (123.9 vs 166.2), and neither alone recovers the joint oracle (61.5).